AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 6 · WORKBOOK

Sound Becomes a Signal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected sound-sample record.

Read timed values, inspect discarded positions, compare summaries and diagnose a speech pipeline.

Keep one habit: Keep your original record, its timing and the reason for each conclusion.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Pictures Become Numbers

Sensors Are Messengers

MAIN LESSON · SECTION 12

W1 · Read samples and gaps

> Count the recorded positions and the spaces between them.

Your evidence record

- Five **sample**s have values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. State the **sample** count, interval and elapsed time from first to last. Explain why there are four gaps.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Inspect what was discarded

> Use W1's complete timed record before retaining a subset.

Your evidence record

- Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original **signal** that these retained values do not justify.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Compare equal summaries

> Explain the pattern information a maximum omits.

Your evidence record

- Sequences A=0,2,0,2,0 and B=0,0,2,0,0 use the same time interval. State their maximum values and one difference the maximum discards. Name information that would preserve that difference.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Check the speech pipeline

> Find the first mismatch before a later action.

Your evidence record

- A recording says Set the timer for fourteen minutes. A **transcript** says forty minutes and the timer is set to 40.

  Name the first mismatch, propose a confirmation step and explain why the **transcript** cannot be treated as a complete copy of the recording.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Construct a new hidden change

> Keep one first attempt before hints or feedback.

Read and explain independently

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Construct a new hidden change

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom **noise** makes that **input** unreliable.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Inspect what was discarded for Sound as **sample**d information. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Sound as **sample**d information. Proposed explanation: “Five **sample**s make five gaps”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A microphone records quiet and loud claps at the same distance and settings.

What **waveform** difference is expected? Does it reveal a spoken word?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 6 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

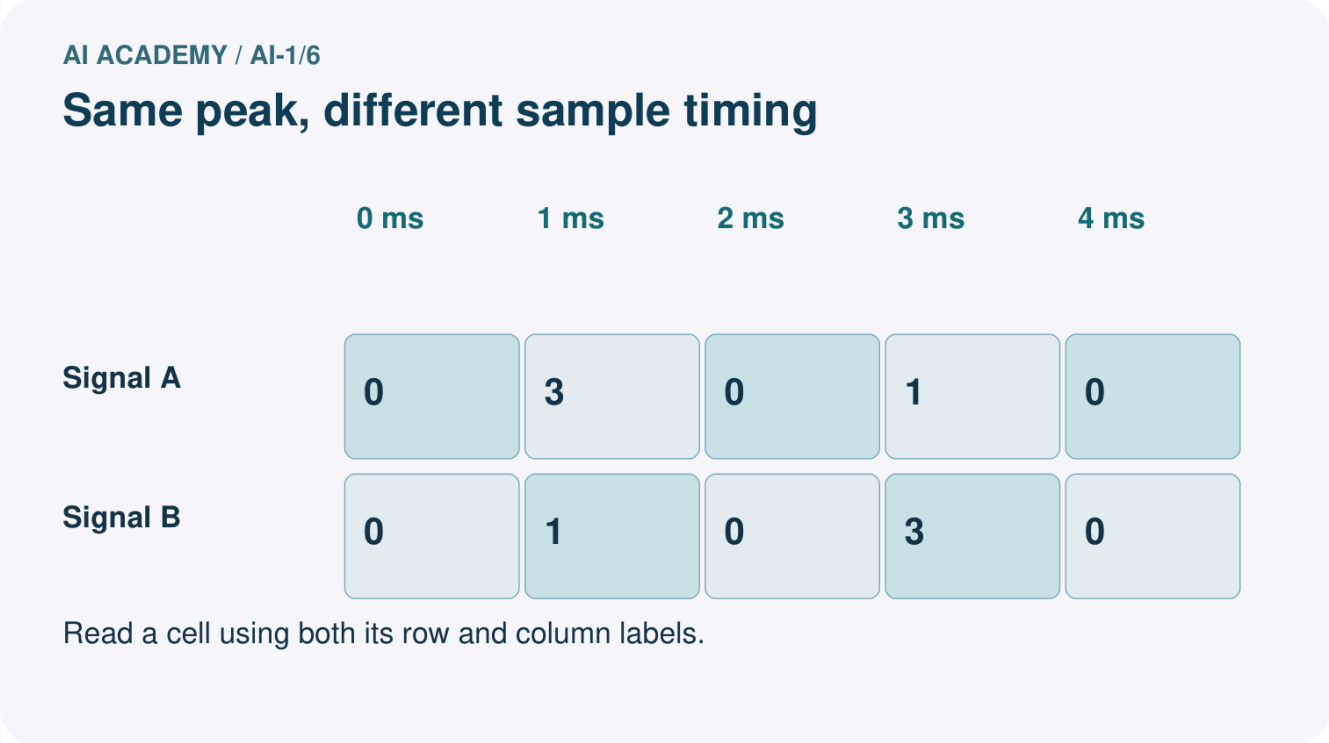
> Keep the explanation and the support you used.

Use this when helpful

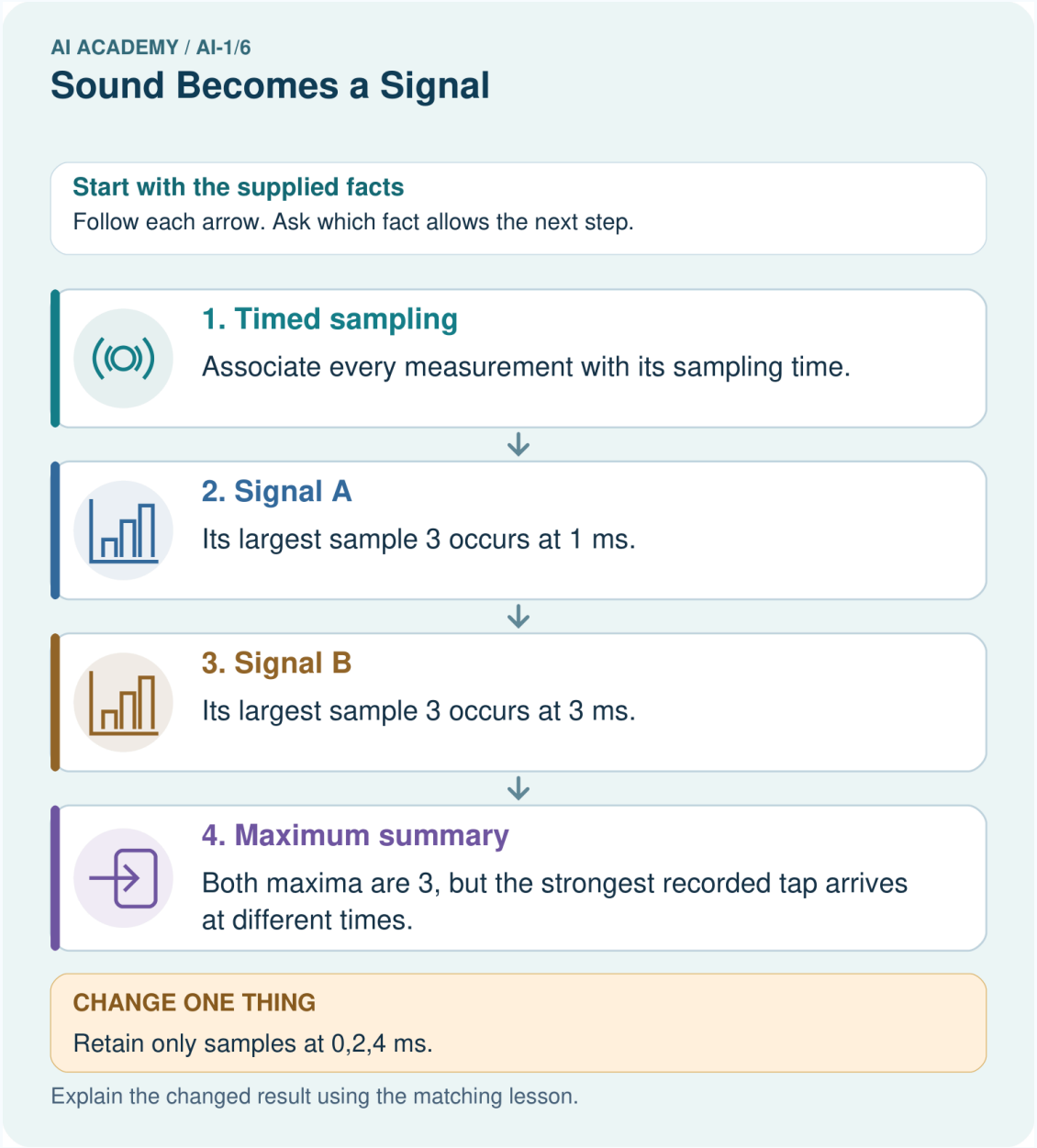
A fictional microphone display records five **sample**s during a short classroom tap.

Times are 0,1,2,3,4 ms in order.

Signal A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Does the all-zero retained record prove the full **signal** was silent? Use the picture.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional microphone display records five **sample**s during a short classroom tap.
- Which labelled step matches this detail? Both maxima are 3, but the strongest recorded tap arrives at different times.
- 1. **Signal** A
- 2. **Signal** B
- 3. Maximum summary
- 4. Timed sampling
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Maximum summary Both maxima are 3, but the strongest recorded tap arrives at different times. The sequences contain the same values in different orders.

Keeping the maximum preserves peak recorded value while losing its timing. A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Changed-case question

- Changed-case practice: Retain only **sample**s at 0,2,4 ms. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Both retained sequences are 0,0,0.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Both retained sequences are 0,0,0. The selected times omit both nonzero **sample**s from each **signal**.

Repair and save your work

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Sound Becomes a Signal: the case

AI ACADEMY / AI-1/6 · 01 · LOOK AT THE FACTS

Sound Becomes a Signal: the case

The situation

A fictional microphone display records five samples during a short classroom tap.

Supplied evidence

Times are 0,1,2,3,4 ms in order.
Signal A: 0,3,0,1,0. Signal B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.

By the end, I can connect ordered sound samples to timing and explain what a maximum loses.

Follow the labelled picture with the matching explanation. Open the vector diagram

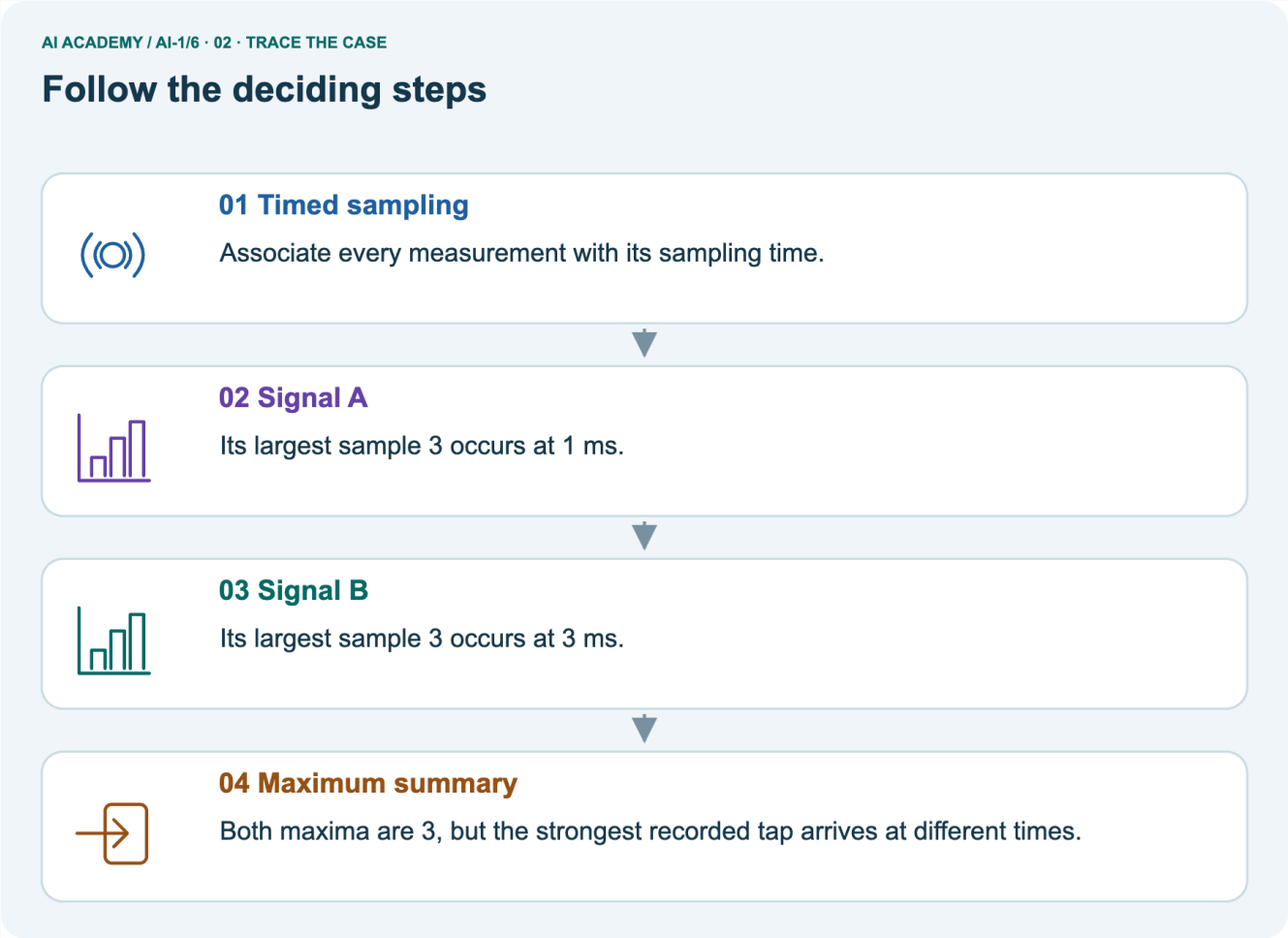
A fictional microphone display records five **sample**s during a short classroom tap. Supplied facts: Times are 0,1,2,3,4 ms in order. **Signal** A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0.

Toy values are signed measurement units; these happen to be nonnegative.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



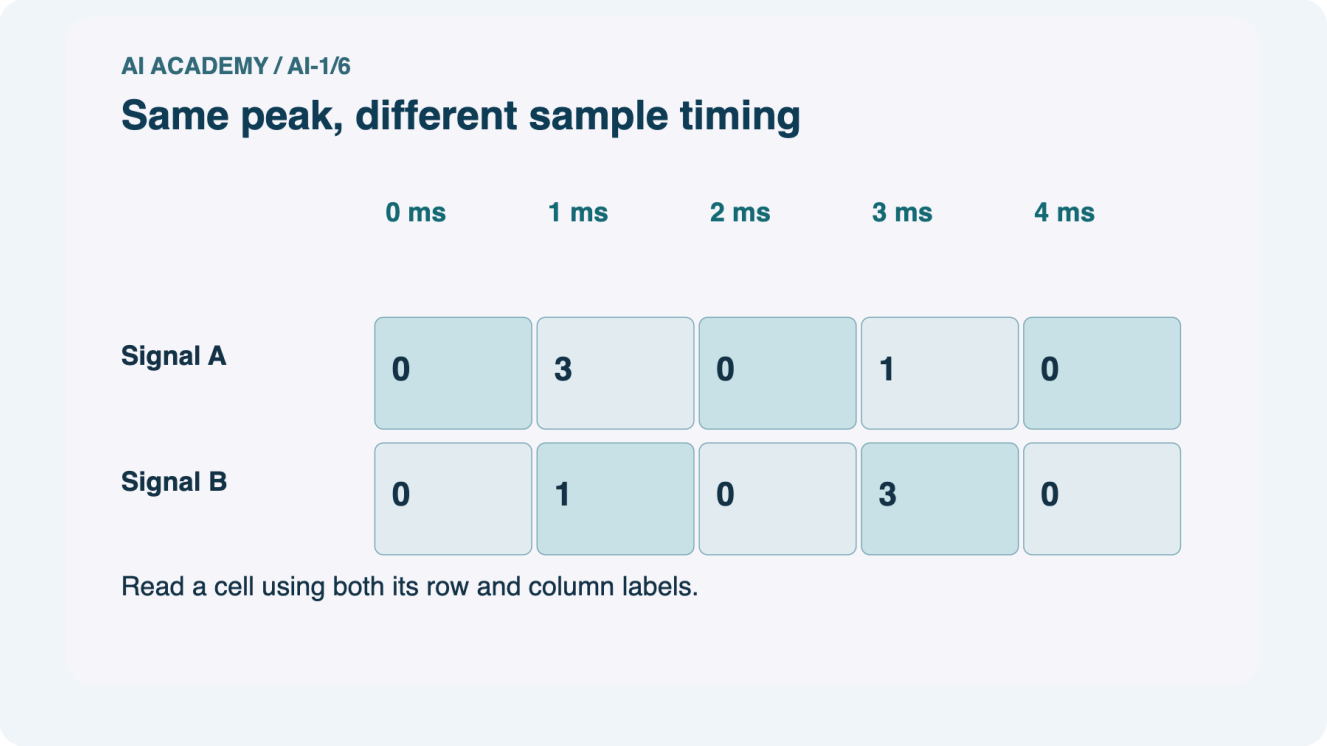
Follow the labelled picture with the matching explanation. Open the vector diagram

Timed sampling: Associate every measurement with its sampling time. → **Signal** A: Its largest **sample** 3 occurs at 1 ms. → **Signal** B: Its largest **sample** 3 occurs at 3 ms. → Maximum summary: Both maxima are 3, but the strongest recorded tap arrives at different times.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Same peak, different sample timing



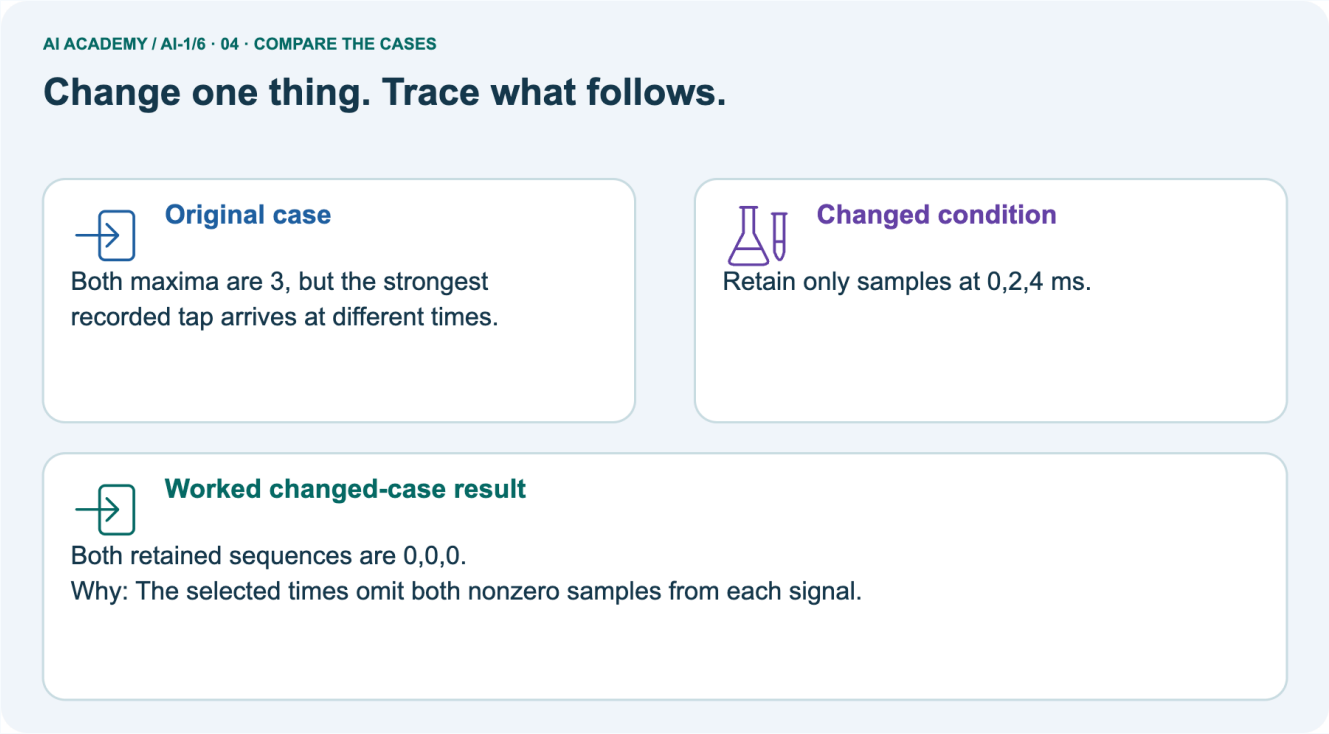
Follow the labelled picture with the matching explanation. Open the vector diagram

Same peak, different **sample** timing. **Signal** A: 0 ms = 0, 1 ms = 3, 2 ms = 0, 3 ms = 1, 4 ms = 0. **Signal** B: 0 ms = 0, 1 ms = 1, 2 ms = 0, 3 ms = 3, 4 ms = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

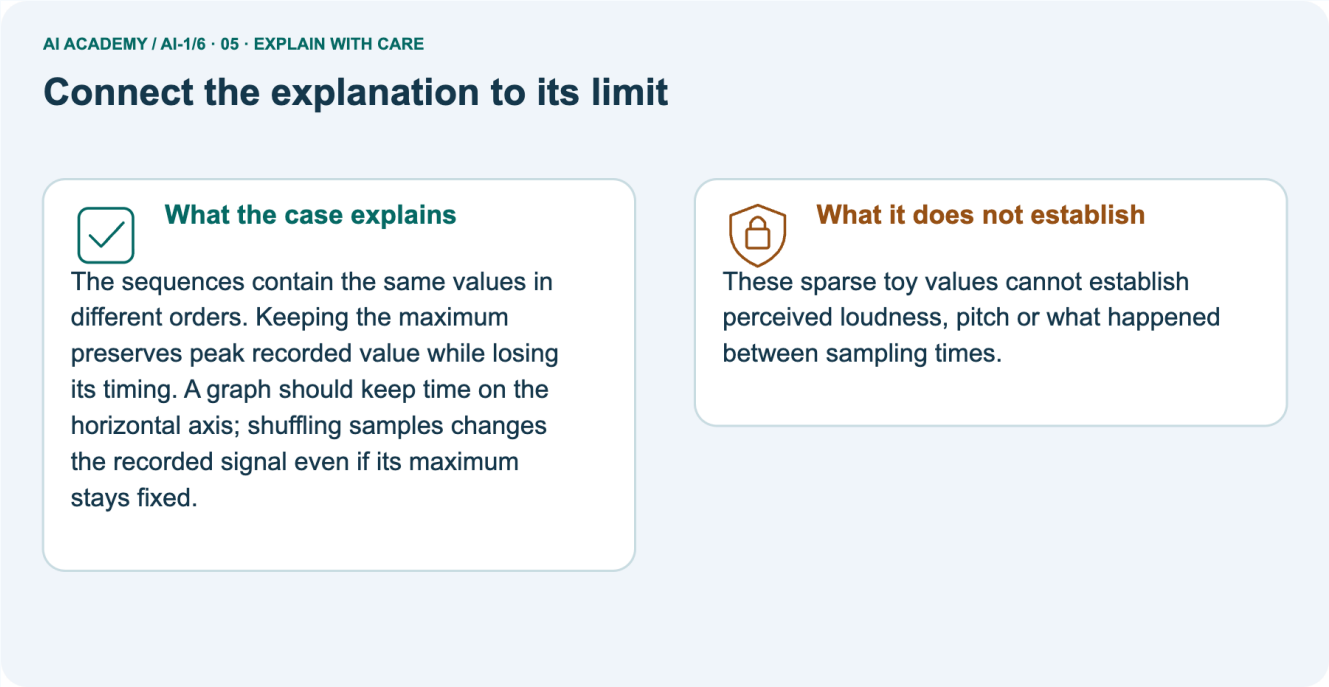
Original case: Both maxima are 3, but the strongest recorded tap arrives at different times. Changed condition: Retain only **sample**s at 0,2,4 ms. Expected result: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero **sample**s from each **signal**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

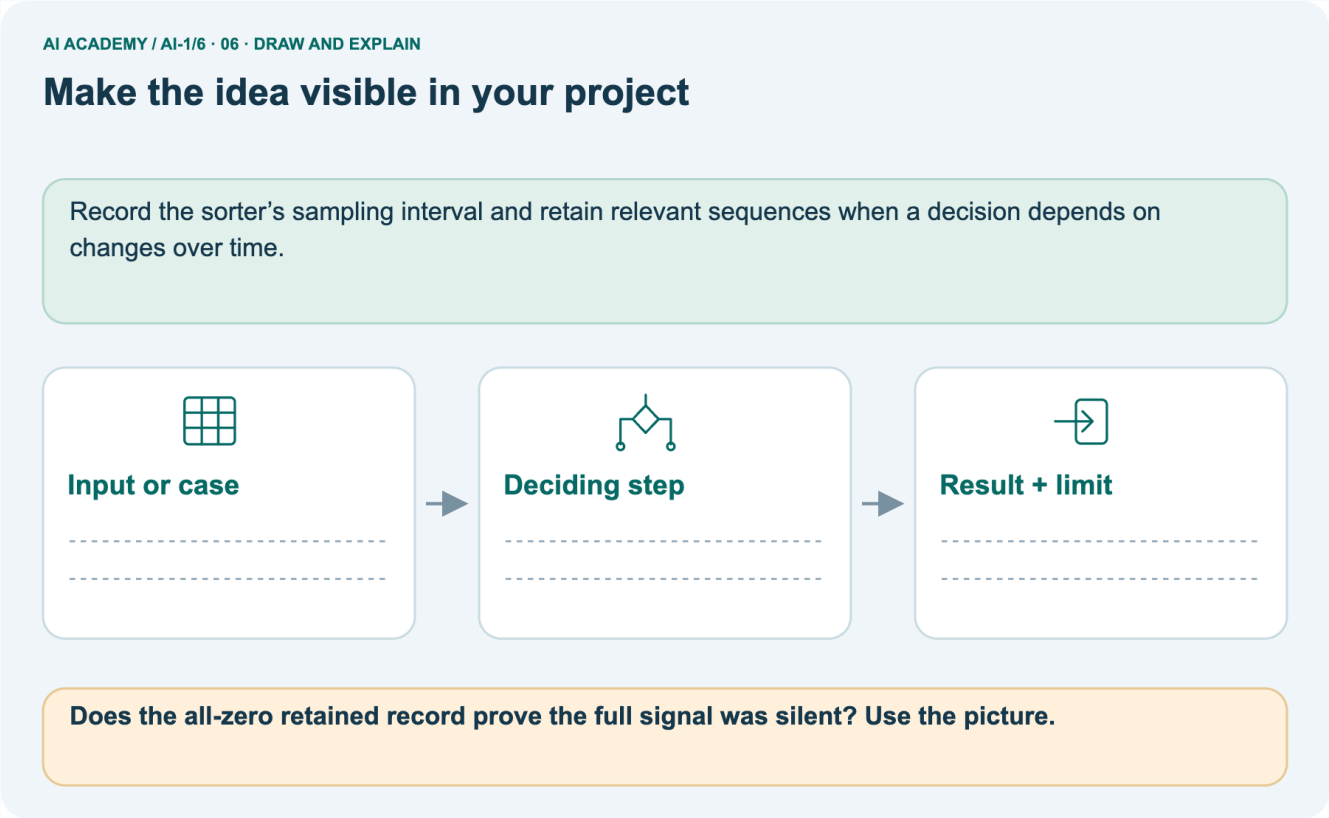
A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Limit: These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time. Draw your **input**, deciding step, result and limit.

Supported question: Does the all-zero retained record prove the full **signal** was silent? Use the picture.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Sound Becomes a Signal: the case

02. Follow the deciding steps

03. Same peak, different sample timing

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.